

1.3 Inverses and generalized inverses

1.3.1 The inverse of a square matrix

An $n \times n$ matrix $\mathbf{A}$ is **nonsingular** if there is a matrix $\mathbf{A}^{-1}$ with $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$. The inverse is unique: if $\mathbf{B}$ and $\mathbf{C}$ both qualify, then $\mathbf{B} = \mathbf{B}(\mathbf{A}\mathbf{C}) = (\mathbf{B}\mathbf{A})\mathbf{C} = \mathbf{C}$.

Proposition 1.3.1

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent: (a) $\mathbf{A}$ is nonsingular; (b) $\text{rank}(\mathbf{A}) = n$; (c) $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$; (d) $\mathbf{B}\mathbf{A} = \mathbf{I}$ for some $\mathbf{B}$; (e) $\mathbf{A}\mathbf{C} = \mathbf{I}$ for some $\mathbf{C}$. When they hold, $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$, and

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top, \quad (\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular $\mathbf{A}, \mathbf{B}$ of the same size. A nonsingular symmetric matrix has a symmetric inverse.

Proof. (b) $\Leftrightarrow$ (c) is rank-nullity. (a) $\Rightarrow$ (d) and (a) $\Rightarrow$ (e) are trivial. (d) $\Rightarrow$ (c): $\mathbf{A}\mathbf{x} = \mathbf{0}$ gives $\mathbf{x} = \mathbf{B}\mathbf{A}\mathbf{x} = \mathbf{0}$. (e) $\Rightarrow$ (b): $\mathbf{I} = \mathbf{A}\mathbf{C}$ has rank n , and $\text{rank}(\mathbf{A}\mathbf{C}) \leq \text{rank}(\mathbf{A})$. (b) $\Rightarrow$ (a): since $\mathcal{C}(\mathbf{A}) = \mathbb{R}^n$, we can solve $\mathbf{A}\mathbf{c}_j = \mathbf{e}_j$ for each j , giving $\mathbf{A}\mathbf{C} = \mathbf{I}$. Then $\mathbf{A}(\mathbf{C}\mathbf{A} - \mathbf{I}) = \mathbf{0}$, and because $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ each column of $\mathbf{C}\mathbf{A} - \mathbf{I}$ is zero. So $\mathbf{C}\mathbf{A} = \mathbf{I}$ as well. If $\mathbf{B}\mathbf{A} = \mathbf{I}$ then $\mathbf{B} = \mathbf{B}\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}$, and similarly for $\mathbf{C}$. The formulas are verified by multiplication: $\mathbf{A}^\top(\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}$ and $(\mathbf{A}\mathbf{B})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$. If $\mathbf{A} = \mathbf{A}^\top$, then $\mathbf{A}^{-1} = (\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$. ■

A rectangular matrix has no two-sided inverse. If $\mathbf{X}$ is $n \times p$ with full column rank, then $\mathbf{X}^\top \mathbf{X}$ is nonsingular by Proposition 1.2.3(c), and $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is a *left inverse*: it multiplies $\mathbf{X}$ on the left to give $\mathbf{I}_p$. This is the matrix that computes least squares coefficients.

1.3.2 Generalized inverses

When $\mathbf{A}$ is singular or rectangular we still want a matrix that inverts $\mathbf{A}$ on the vectors where inversion makes sense, namely those in $\mathcal{C}(\mathbf{A})$. The following weak requirement turns out to be exactly what solving equations needs.

Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is any $n \times m$ matrix $\mathbf{A}^-$ with

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that $\mathbf{A}\mathbf{A}^-$ acts as the identity on every column of $\mathbf{A}$, hence on all of $\mathcal{C}(\mathbf{A})$: if $\mathbf{u} = \mathbf{A}\mathbf{b}$ then $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{A}\mathbf{A}^-\mathbf{A}\mathbf{b} = \mathbf{u}$. So $\mathbf{A}^-\mathbf{u}$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{u}$ whenever one exists. If $\mathbf{A}$ is nonsingular, multiplying the definition by $\mathbf{A}^{-1}$ on both sides gives

$\mathbf{A}^- = \mathbf{A}^{-1}$. In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A}_{IJ}$ is the nonsingular $r \times r$ submatrix of Proposition 1.2.4, the $n \times m$ matrix $\mathbf{G}$ that holds $\mathbf{A}_{IJ}^{-1}$ in the rows indexed by J and the columns indexed by I , and zeros elsewhere, is a generalized inverse of $\mathbf{A}$.

Proof. For $\mathbf{A} = \mathbf{0}$ any matrix of the right size works. Otherwise let $\mathbf{E}_I$ and $\mathbf{E}_J$ be the columns of $\mathbf{I}_m$ and $\mathbf{I}_n$ indexed by I and J . Then $\mathbf{G} = \mathbf{E}_J \mathbf{A}_{IJ}^{-1} \mathbf{E}_I^\top$, $\mathbf{A} \mathbf{E}_J = \mathbf{A}_{.J}$ and $\mathbf{E}_I^\top \mathbf{A} = \mathbf{A}_{I.}$. So $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.} = \mathbf{A}$ by Eq. 1.2.1. ■

Proposition 1.3.3 — Properties of generalized inverses

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$.

- (a) $\mathbf{A} \mathbf{G}$ and $\mathbf{G} \mathbf{A}$ are idempotent. Moreover $\mathcal{C}(\mathbf{A} \mathbf{G}) = \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{G} \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A} \mathbf{G}) = \text{rank}(\mathbf{G} \mathbf{A}) = r \leq \text{rank}(\mathbf{G})$.
- (b) $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top$.
- (c) If $\mathbf{G}_1, \mathbf{G}_2$ are generalized inverses of $\mathbf{A}$, so is $\mathbf{G}_1 \mathbf{A} \mathbf{G}_2$. In particular $\mathbf{G} \mathbf{A} \mathbf{G}$ is a generalized inverse which also satisfies $(\mathbf{G} \mathbf{A} \mathbf{G}) \mathbf{A} (\mathbf{G} \mathbf{A} \mathbf{G}) = \mathbf{G} \mathbf{A} \mathbf{G}$.
- (d) If $\mathbf{A}$ is symmetric, then $\mathbf{G}^\top$ and $\frac{1}{2}(\mathbf{G} + \mathbf{G}^\top)$ are generalized inverses of $\mathbf{A}$, so a symmetric generalized inverse always exists.
- (e) $\mathbf{G} + (\mathbf{I} - \mathbf{G} \mathbf{A}) \mathbf{U} + \mathbf{V} (\mathbf{I} - \mathbf{A} \mathbf{G})$ is a generalized inverse of $\mathbf{A}$ for every $\mathbf{U}$ and $\mathbf{V}$ of size $n \times m$.

Proof.

- (a) $(\mathbf{A} \mathbf{G})^2 = (\mathbf{A} \mathbf{G} \mathbf{A}) \mathbf{G} = \mathbf{A} \mathbf{G}$, and likewise for $\mathbf{G} \mathbf{A}$. Next, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{A} \mathbf{G} \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A} \mathbf{G}) \subseteq \mathcal{C}(\mathbf{A})$, and $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{G} \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A} \mathbf{G} \mathbf{A}) = \mathcal{N}(\mathbf{A})$. The ranks follow from these and rank-nullity, and $r = \text{rank}(\mathbf{A} \mathbf{G}) \leq \text{rank}(\mathbf{G})$.
- (b) Transpose $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}$. (c) $\mathbf{A} \mathbf{G}_1 \mathbf{A} \mathbf{G}_2 \mathbf{A} = \mathbf{A} \mathbf{G}_2 \mathbf{A} = \mathbf{A}$. The second statement is $\mathbf{G} \mathbf{A} \mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{G} (\mathbf{A} \mathbf{G} \mathbf{A}) \mathbf{G} = \mathbf{G} \mathbf{A} \mathbf{G}$.
- (c) By (b), $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top = \mathbf{A}$. The condition $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}$ is linear in $\mathbf{G}$, so it is preserved by averaging.
- (d) Multiply out, using $\mathbf{A} (\mathbf{I} - \mathbf{G} \mathbf{A}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{A} \mathbf{G}) \mathbf{A} = \mathbf{0}$. ■

A generalized inverse with the extra property in (c), $\mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{G}$, is called **reflexive**. Exercise (B1) shows that the reflexive ones are exactly those of rank r . Part (e) in fact produces *every* generalized inverse from a single one as $\mathbf{U}$ and $\mathbf{V}$ vary. Idempotence in (a) means that $\mathbf{A} \mathbf{G}$ and $\mathbf{G} \mathbf{A}$ are projections, generally not orthogonal ones. Chapter 6 shows that the particular

combination $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$ is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$.

1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of $\mathbf{A}$ is the matrix $\mathbf{A}^+$ satisfying

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+, \quad (\mathbf{A}\mathbf{A}^+)^\top = \mathbf{A}\mathbf{A}^+, \quad (\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If $\mathbf{G}$ and $\mathbf{H}$ both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}(\mathbf{A}\mathbf{G})^\top = \mathbf{G}\mathbf{G}^\top(\mathbf{A}\mathbf{H}\mathbf{A})^\top = \mathbf{G}(\mathbf{A}\mathbf{G})^\top(\mathbf{A}\mathbf{H})^\top = \mathbf{G}\mathbf{A}\mathbf{G}\mathbf{A}\mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H},$$

where the steps use the second and third conditions for $\mathbf{G}$, the first for $\mathbf{H}$, and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{H}\mathbf{A})^\top \mathbf{H} = \mathbf{A}^\top \mathbf{H}^\top \mathbf{H} = (\mathbf{A}\mathbf{G}\mathbf{A})^\top \mathbf{H}^\top \mathbf{H} = (\mathbf{G}\mathbf{A})^\top (\mathbf{H}\mathbf{A})^\top \mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H}\mathbf{A}\mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H},$$

so $\mathbf{G} = \mathbf{H}$. Existence follows from the singular value decomposition, which gives $\mathbf{A}^+$ explicitly (Theorem 1.8.1).